

Design and Simulation of an IoT-Based Cyclone Alert System

Using Cisco Packet Tracer

Project Paper — IoT Network Simulation

1. Title of the Project

Design and Simulation of an IoT-Based Cyclone Alert System (CycloneAlert) Using Cisco Packet Tracer.

2. Objective

The objective of this project is to design and simulate an Internet of Things (IoT) based early-warning system that can automatically detect the environmental conditions associated with a cyclone — rising temperature, high humidity, and strong wind — and trigger an audible siren alert without human intervention. The project demonstrates how IoT sensors, a central home gateway, and rule-based automation can be combined in Cisco Packet Tracer to model a real-world disaster-warning network.

3. Problem Statement

Cyclones are natural disasters that develop rapidly and can cause severe loss of life and property, especially in coastal regions. A major challenge in disaster management is the timely detection and communication of hazardous weather conditions to the people at risk. Manual monitoring of weather parameters is slow, error-prone, and often unavailable in remote areas. There is a need for an automated, low-cost, and reliable system that continuously monitors key environmental parameters and instantly raises an alarm when conditions matching a cyclone are detected, giving people valuable time to seek safety.

4. Components Used

The following IoT components, available in the Cisco Packet Tracer IoT device library, were used to build and simulate the CycloneAlert system. Certain generic IoT devices were repurposed and relabeled to represent the required environmental sensors:

- Home Gateway (DLC100) — acts as the central IoT hub / access point that wirelessly connects all sensor and actuator devices and hosts the IoT Server with the Device Conditions (rule) engine.
- Lawn Sprinkler device, relabeled “Temperature” — represents the temperature sensor of the system.
- Fan device, relabeled “Wind” — represents the wind-speed sensor of the system.
- Humidifier device, relabeled “Humidifier” — represents the humidity sensor of the system.

- Siren — the actuator/alarm device that produces the audible cyclone warning.
- Smartphone (Smartphone0, running the SMARTPHONE-PT app) — connects to the Home Gateway to allow remote monitoring and control of the connected devices.

All devices communicate wirelessly with the Home Gateway, which in turn hosts the logic that evaluates sensor states and drives the siren.

5. System Architecture / Network Diagram

The network follows a star (hub-and-spoke) topology. The Home Gateway (DLC100) sits at the centre of the network and maintains a direct wireless link with every other node — the Temperature sensor (Lawn Sprinkler), the Wind sensor (Fan), the Humidity sensor (Humidifier), the Siren, and the monitoring Smartphone. All sensor readings are reported to the Home Gateway, which evaluates them against a stored rule and, when the rule is satisfied, sends a command to switch the Siren on. The Smartphone app is connected to the same gateway purely for monitoring/control and does not participate in the alert logic.

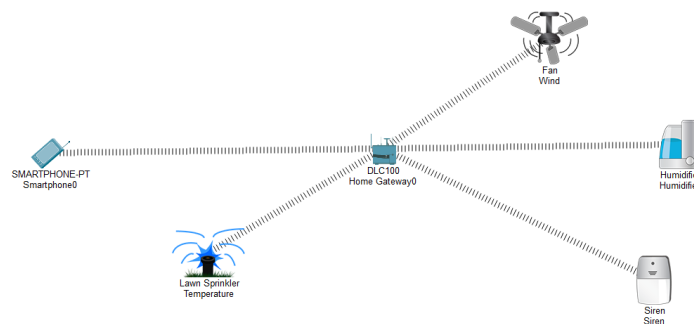


Figure 1: Logical network topology of the CycloneAlert IoT simulation in Cisco Packet Tracer.

6. Working Principle

Each sensor device continuously reports its status to the IoT Server hosted on the Home Gateway:

- The Temperature sensor reports a boolean “Temperature Status” that becomes true once the monitored temperature crosses the configured threshold.
- The Wind sensor reports a “Wind Status” that is classified as Low / Normal / High depending on the simulated wind speed.
- The Humidity sensor reports a “Humidifier Status” that becomes true once the monitored humidity crosses the configured threshold.

The Home Gateway's rule engine (Device Conditions) constantly checks these three status values. Only when all three conditions are simultaneously satisfied — high temperature, high wind, and high humidity, the

combined signature of an approaching cyclone — does the gateway issue an action command that sets the Siren's “On” state to true, sounding the alarm. If any one of the three conditions is not met, the siren remains off, which prevents false alarms triggered by a single abnormal reading.

7. Automation Rules

The alert logic is implemented as a single IoT Server rule named “CycloneAlert”, configured under Conditions → Edit Rule in Packet Tracer's IoT Server interface, as shown below.

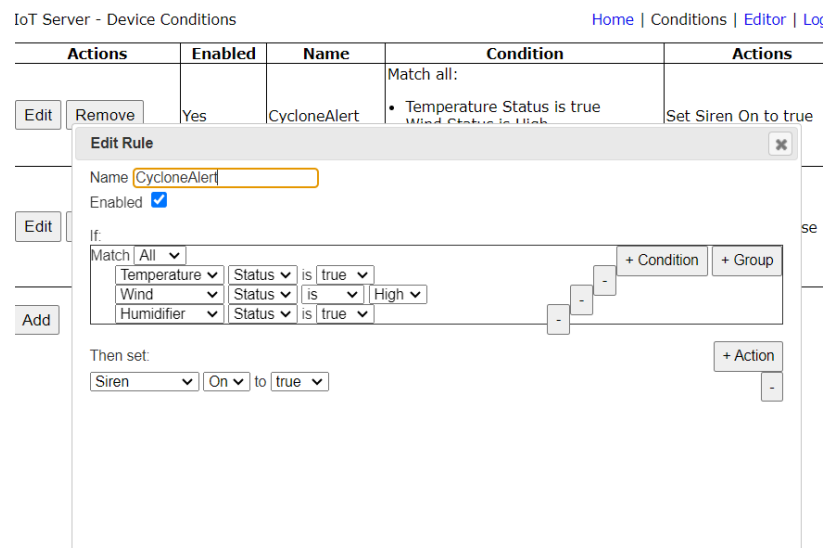


Figure 2: “CycloneAlert” rule configuration in the IoT Server — Device Conditions editor.

Rule definition:

- Match: All of the following conditions (logical AND)
- Condition 1: Temperature – Status is true
- Condition 2: Wind – Status is High
- Condition 3: Humidifier – Status is true
- Action: Set Siren – On to true

Because the rule uses “Match All”, the siren is activated strictly on the combined condition of high temperature, high wind speed, and high humidity, which together model the onset of cyclone-like weather. This AND-based logic was deliberately chosen over an OR-based one to make the alert specific to cyclone conditions rather than to any single weather anomaly.

8. Screenshots of the Simulation

The two key views captured from the Cisco Packet Tracer simulation are the logical network topology (Figure 1, Section 5) showing the Home Gateway and its connected sensor/actuator devices, and the Device

Conditions rule editor (Figure 2, Section 7) showing the configured “CycloneAlert” automation rule. Together these confirm that the physical/logical wireless connectivity and the automation logic were both correctly implemented in the simulated network.

9. Results and Discussion

During simulation, when the Temperature and Humidifier (humidity) devices were driven into their “true” (threshold-exceeded) state and the Fan (wind) device reported a “High” wind status at the same time, the IoT Server evaluated the “CycloneAlert” rule as satisfied and automatically issued the “Set Siren On to true” action, causing the Siren device to activate. When any one of the three parameters did not meet its threshold, the rule condition failed and the siren correctly remained off, confirming that the automation logic responds only to the intended combined cyclone signature and not to isolated sensor fluctuations. The Smartphone app connected to the Home Gateway was able to reflect the live device states, showing that the system also supports remote monitoring in addition to fully automated response.

10. Conclusion

The CycloneAlert project successfully demonstrates how a low-cost IoT network, modelled entirely within Cisco Packet Tracer, can automate the detection of cyclone-prone weather conditions and provide an immediate, human-independent siren warning. By combining three environmental indicators — temperature, humidity, and wind — through a simple AND-based automation rule, the system minimises false alarms while ensuring a timely response when genuine cyclone conditions are detected. This simulation validates the core architecture and logic of the system and can serve as a foundation for a real hardware implementation using actual temperature, humidity, and wind-speed sensors connected to a microcontroller-based gateway.

11. Certificate

